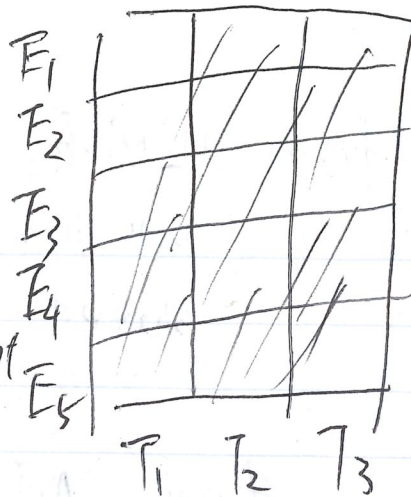


⇒ global routing via opt.

⇓  
Global decide expert assignment



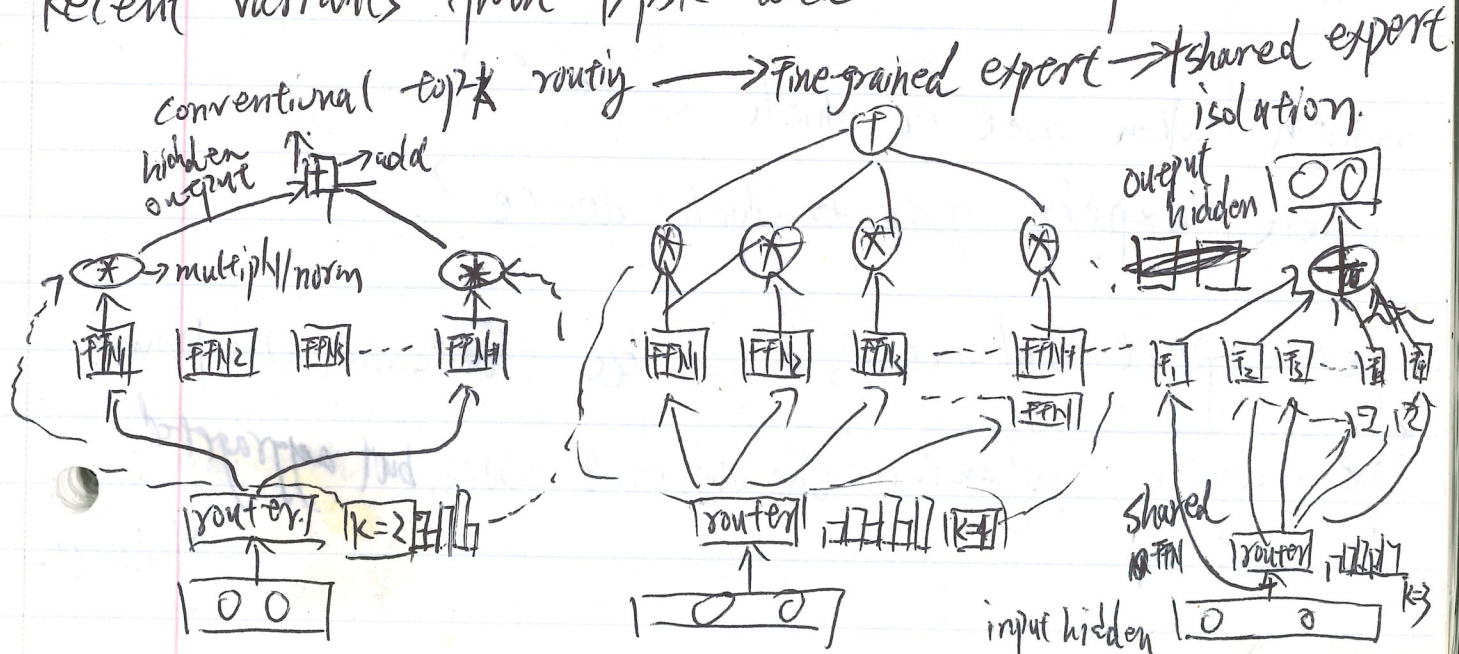
other routing methods:

⇓  
RL to learn routes: used in some earliest work  
 { solve a matching problem: linear assignment for routing = popular

know ~~no~~ LLMs used classical topic routing: how does it work?

⇒ softmax function to output the selection  
 { gated selection by a logistic regressor

Recent variants from Dpsk and other open LLMs





Ablations from OLMoE:

gains from fine-grained experts, none from shared experts

①: expert selection is non-smooth/discrete decision  
②: no gradient flows discrete choice

How do we train MoEs?

Major challenge: we need sparsity for training-time efficiency

But sparse gating decisions are not differentiable!  
Can't be directly optimized  $\Leftarrow J(g(x)) = \text{TopK}(Wx)$ : discrete function; gradients are zero everywhere

Solutions: 1. RL to optimize gating policies

2. stochastic perturbations/approximation

3. Heuristic 'balancing' losses

approximations or relaxations that let training work

which token goes to which expert?

which expert goes to which device?

per-expert balancing: same as the switch transformer

per-device balancing: the objective above, but aggregated by device